Review for the Second Midterm Exam

1. Second Midterm Exam
   - on Friday 13 November, in class or take home
2. One Picture of Each Lecture
   - point location
   - Voronoi diagrams
   - arrangements and duality
   - Delaunay triangulations
   - windowing queries
3. Fall 2023 Second Midterm Exam
   - five questions

MCS 481 Lecture 34
Computational Geometry
Jan Verschelde, 11 November 2026

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the second midterm exam

The exam covers the chapters six to ten:

1. point location,
2. Voronoi diagrams,
3. arrangements and duality,
4. Delaunay triangulations,
5. windowing queries.

There will be five questions, one of each chapter.

The review aims to prepare for the midterm exam,
not to offer a complete coverage of all topics.

The exam is open book, open notes, and you may use your laptop
so you do not have to print out all the slides.

The take-home version of the exam will involve computer experiments;
is due on Monday 16 November, at noon.

skipping policy

About the policies in the course:

- Students may opt out of the exams and
  work on a computer project of their own research interest.
- In particular, concerning the exams:

  If an exam is missed, then greater weight will be placed
  on the final exam,
  especially on the material covered on the missing exam.

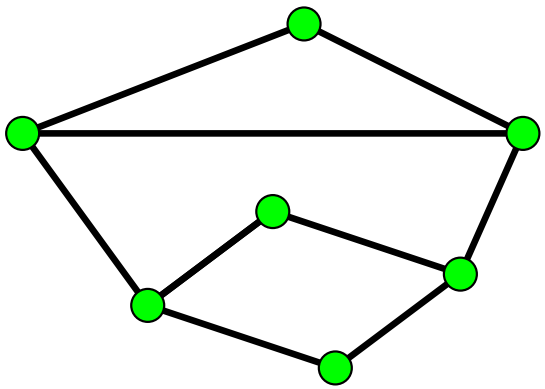
  Students may replace the final exam with a research computer
  project combined with a written technical report.

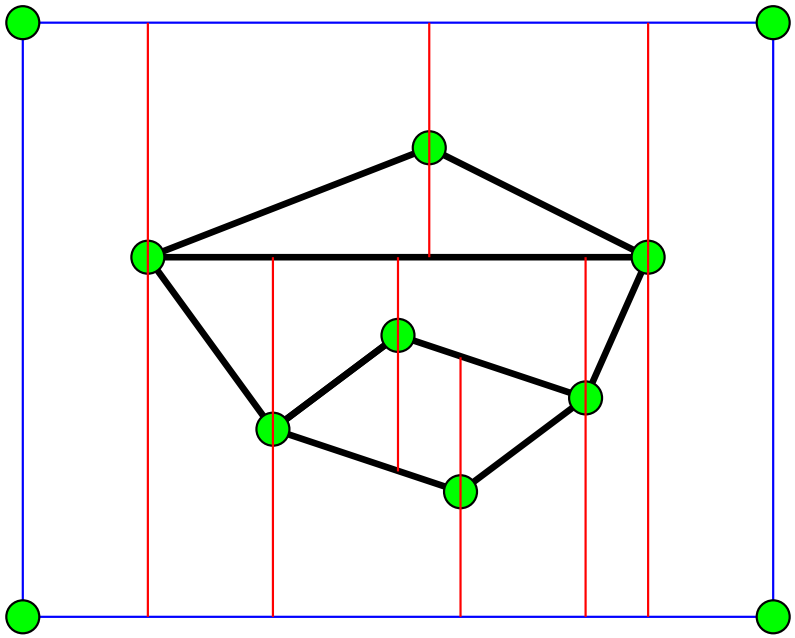
Even if you would opt for a final project,
then taking the exam will give you homework credit.

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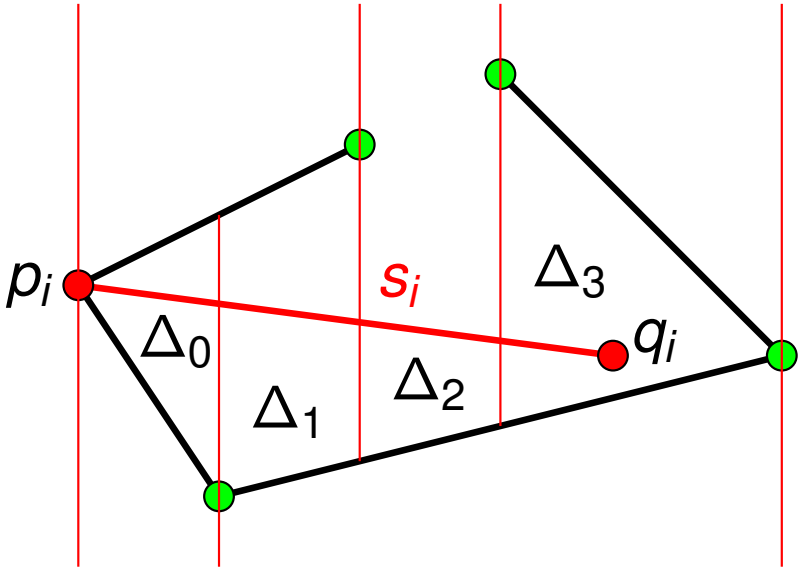
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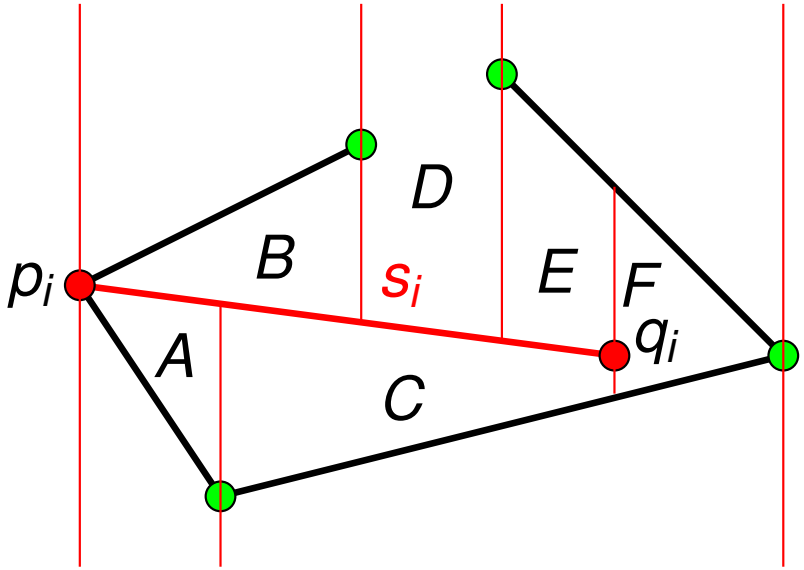
point location and trapezoidal maps

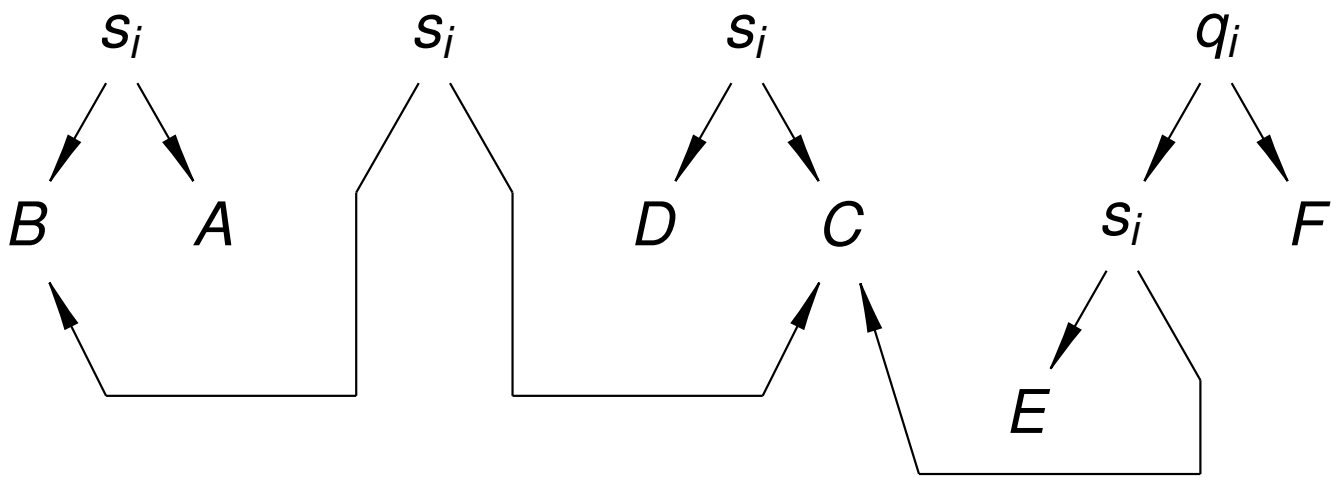




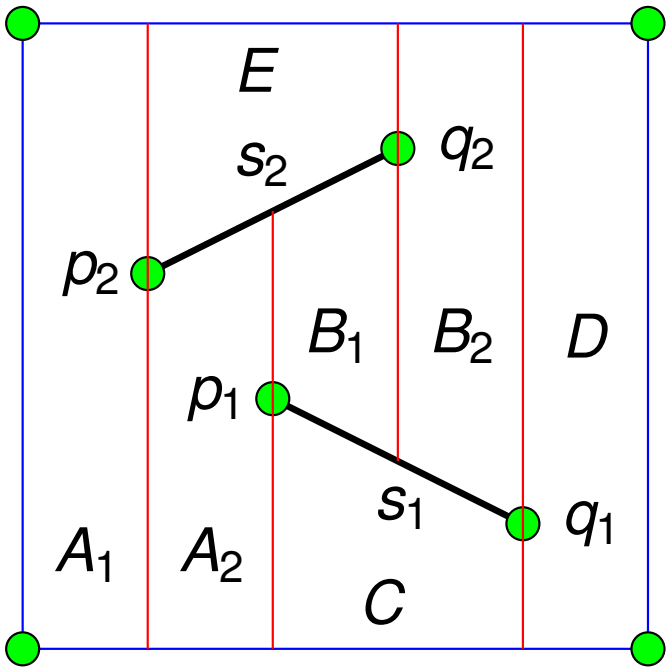
a randomized incremental algorithm

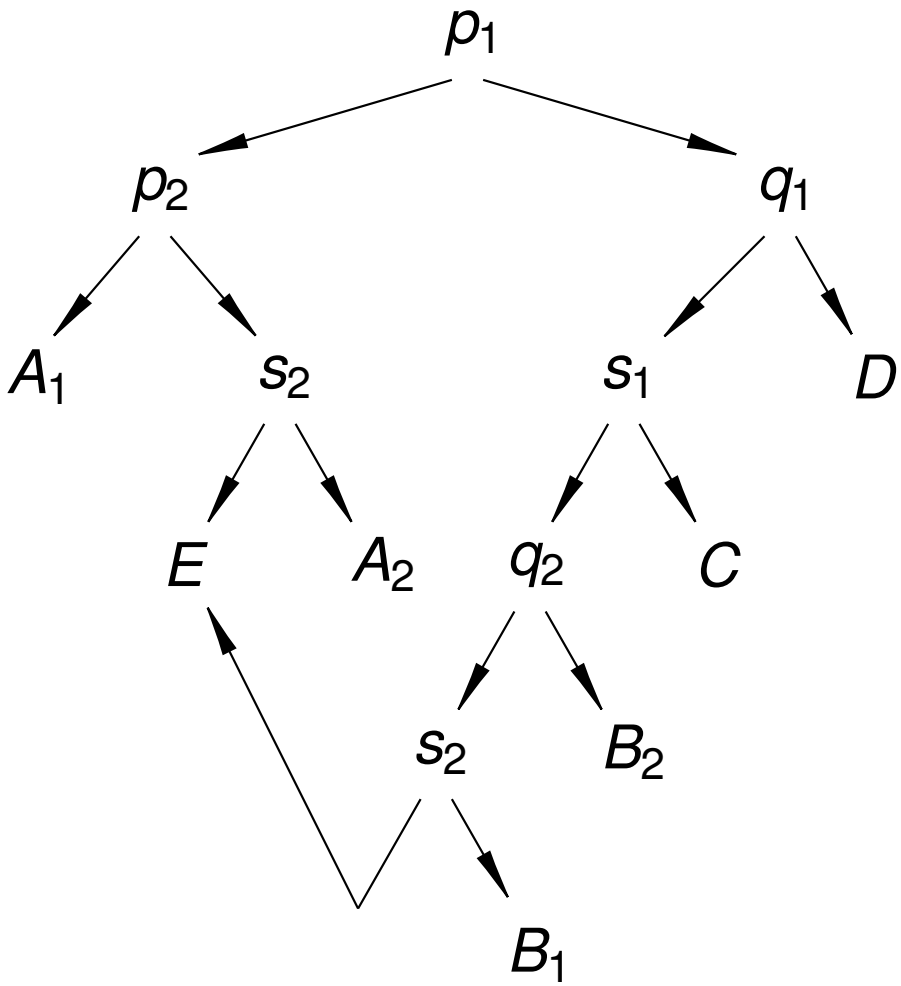






expected cost and degenerate cases





1. randomization to contruct a trapezoidal map

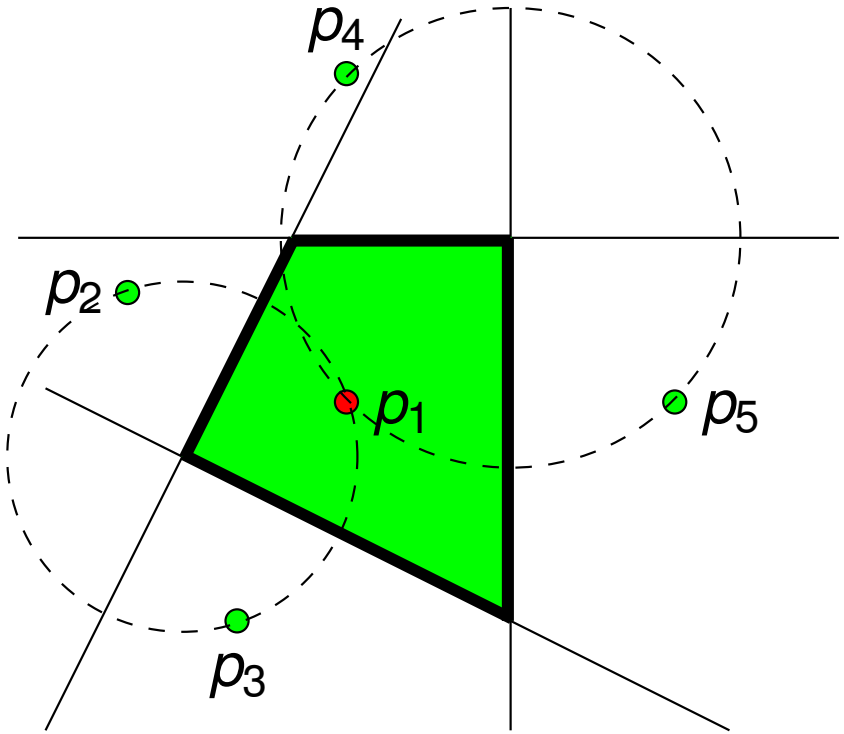
Why is randomization needed to construct a trapezoidal map?

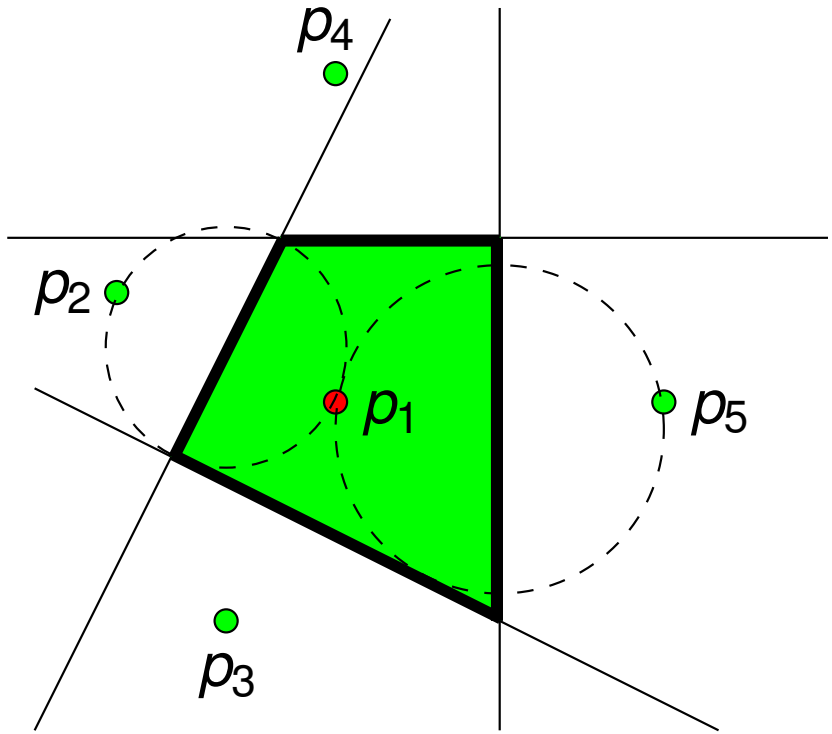
Give a good example to illustrate the need for randomization.

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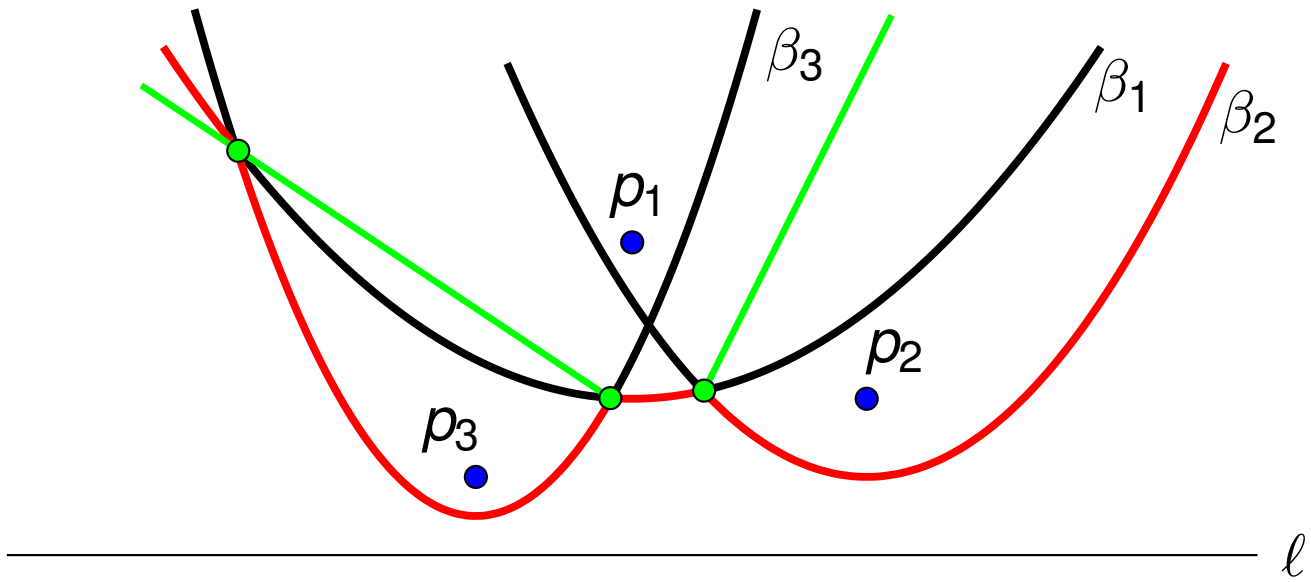
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Voronoi diagrams

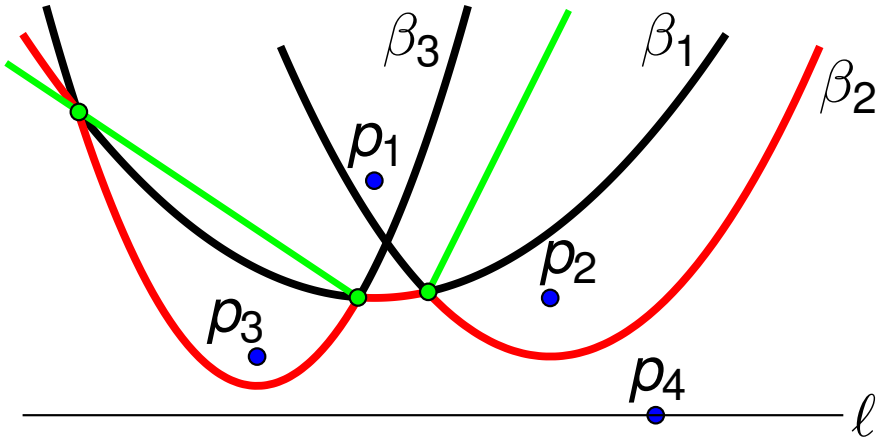


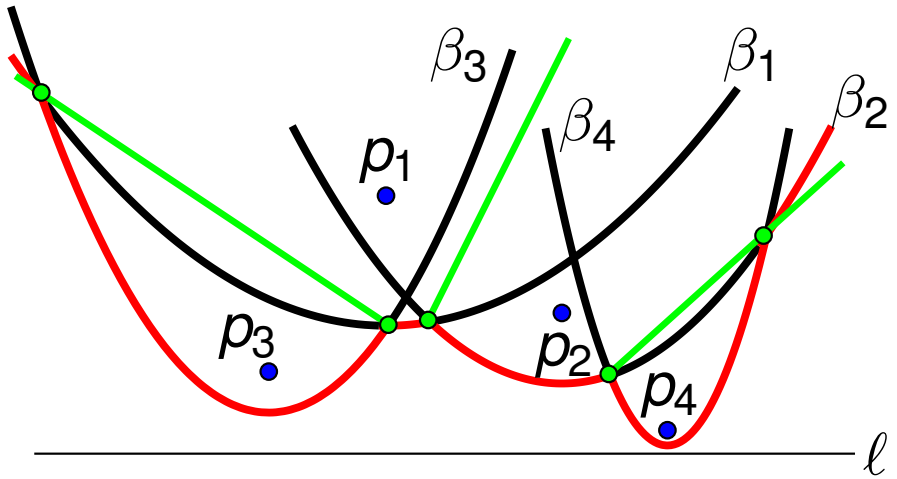


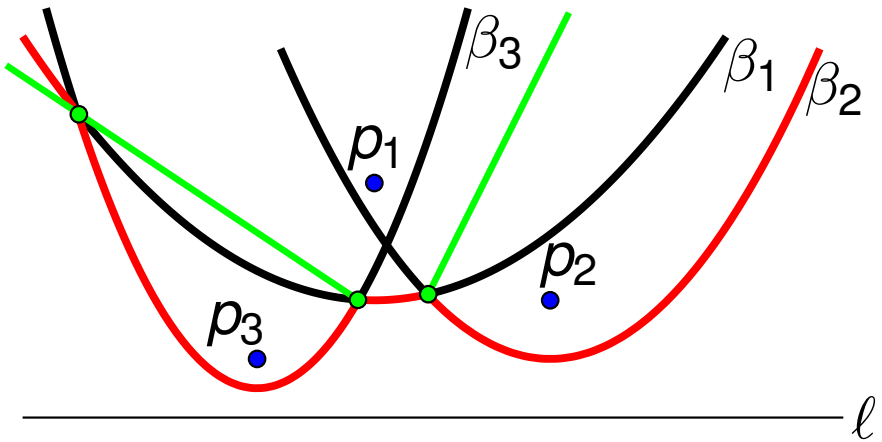
computing Voronoi diagrams

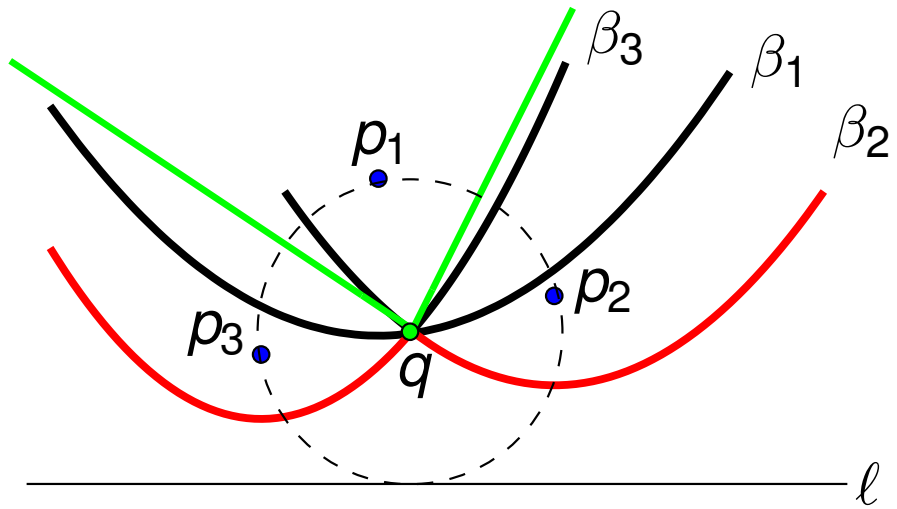


algorithmic cost of Voronoi diagrams









2. the beach line in computing the Voronoi diagram

Consider the set of points

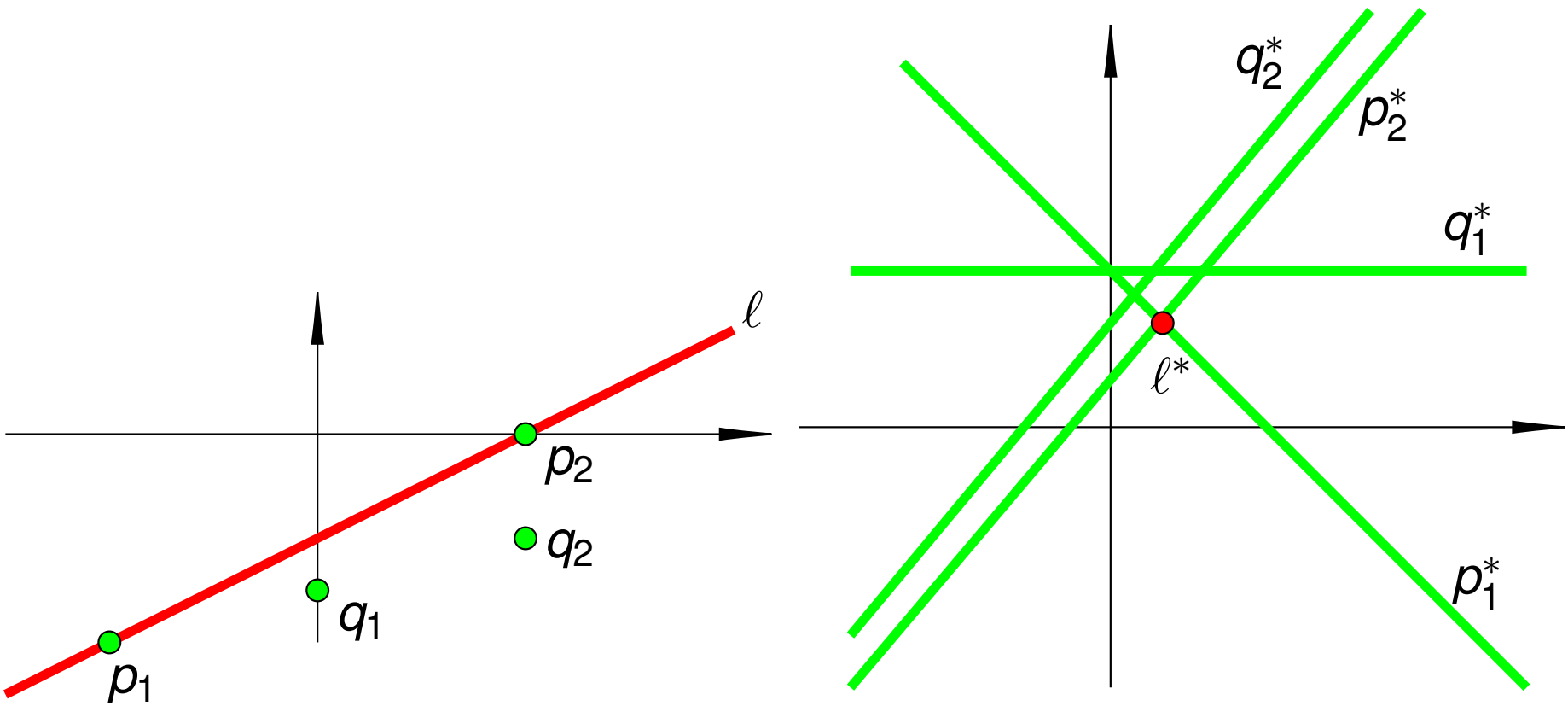
$$S = \{ (86, 74), (27, 88), (88, 70), (92, 50), (41, 60) \}.$$

Draw the evolution of the beach line in the construction of a Voronoi diagram for S.

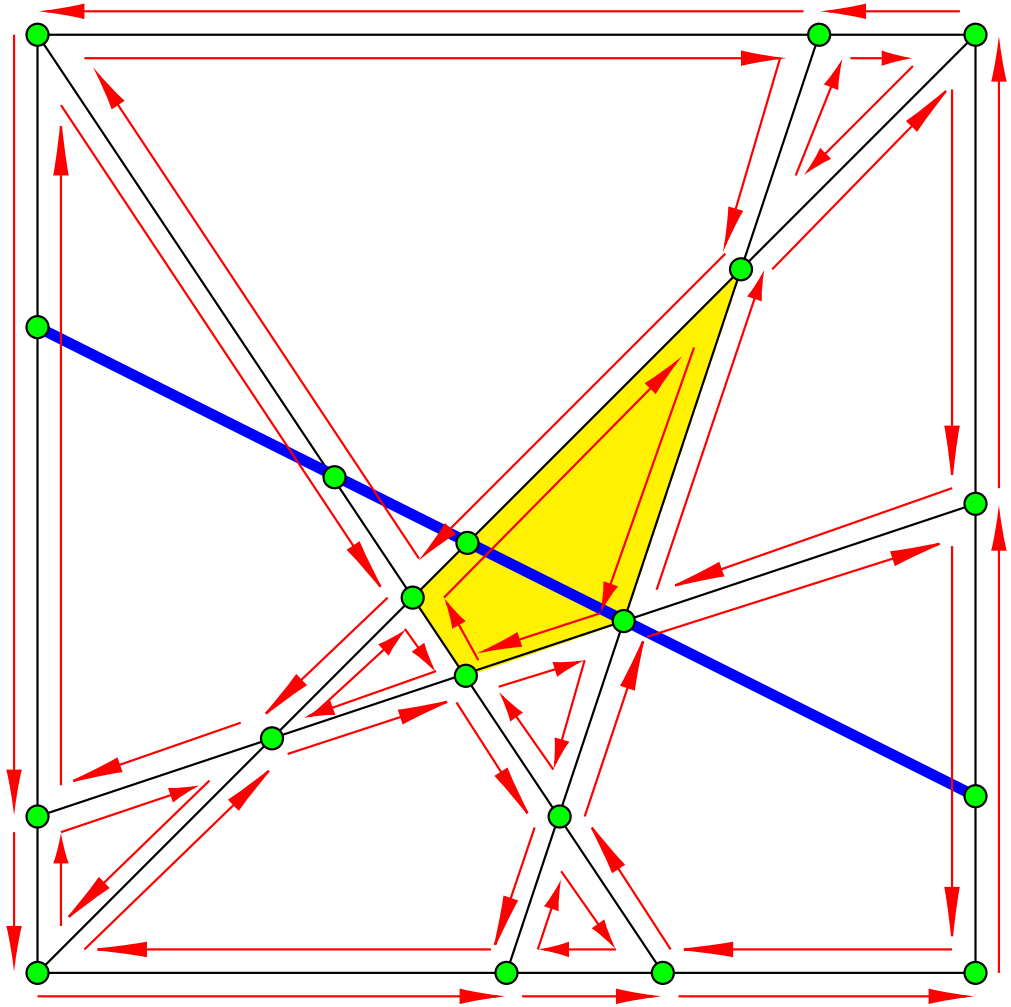
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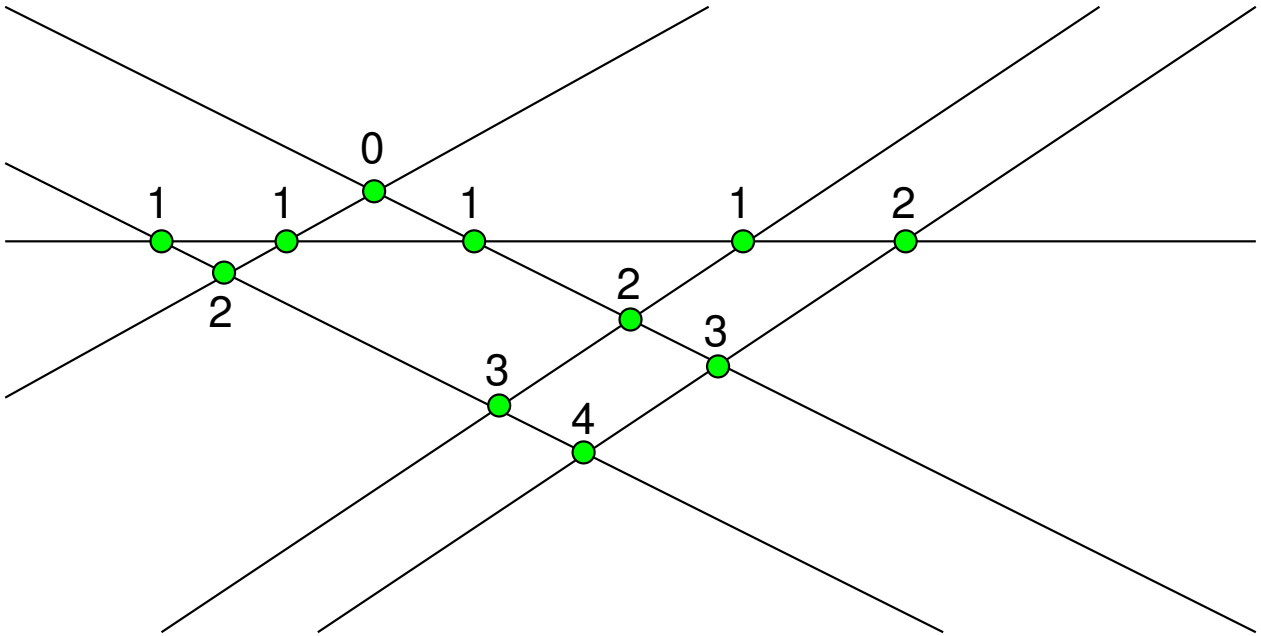
discrepancy and duality



arrangements of lines



the zone theorem



3. constructing an arrangement of lines

In the incremental algorithm to construct an arrangement of lines, we intersect a face with a line.

Consider a subroutine SPLITFACE which takes on input

1. A, an arrangement of lines,
2. f, a face in the arrangement A, and
3. L, a line intersecting f.

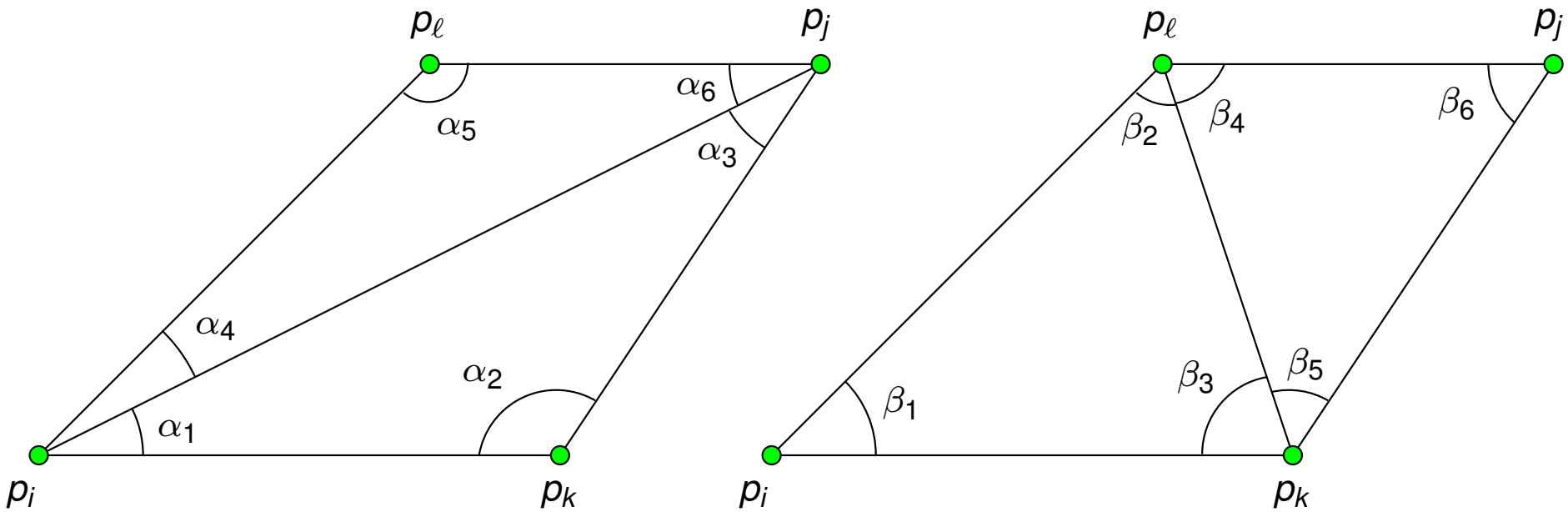
The subroutine splits the face f, adding edges and vertices to A.

Write pseudo code for SPLITFACE.

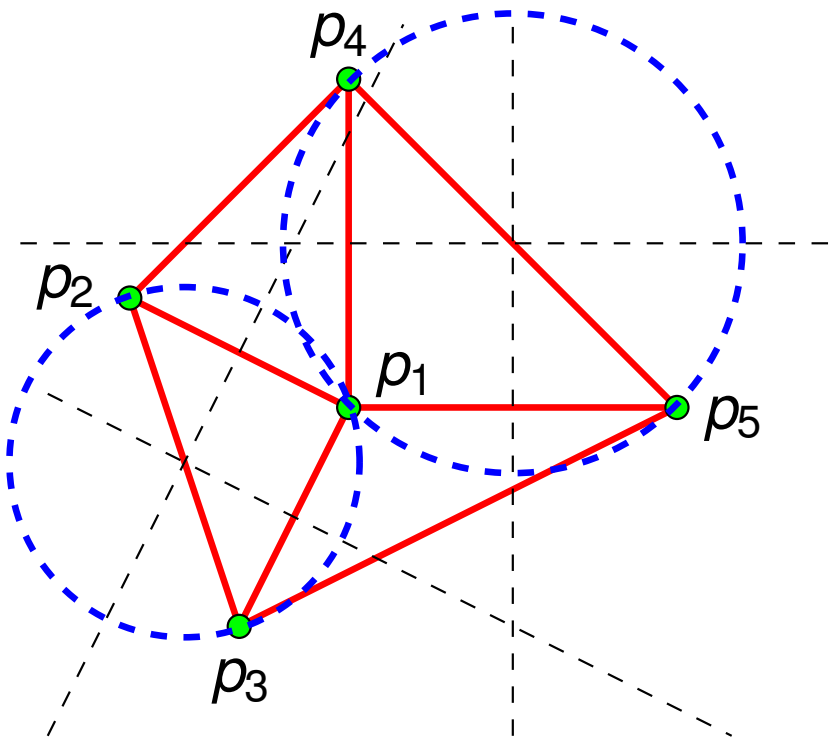
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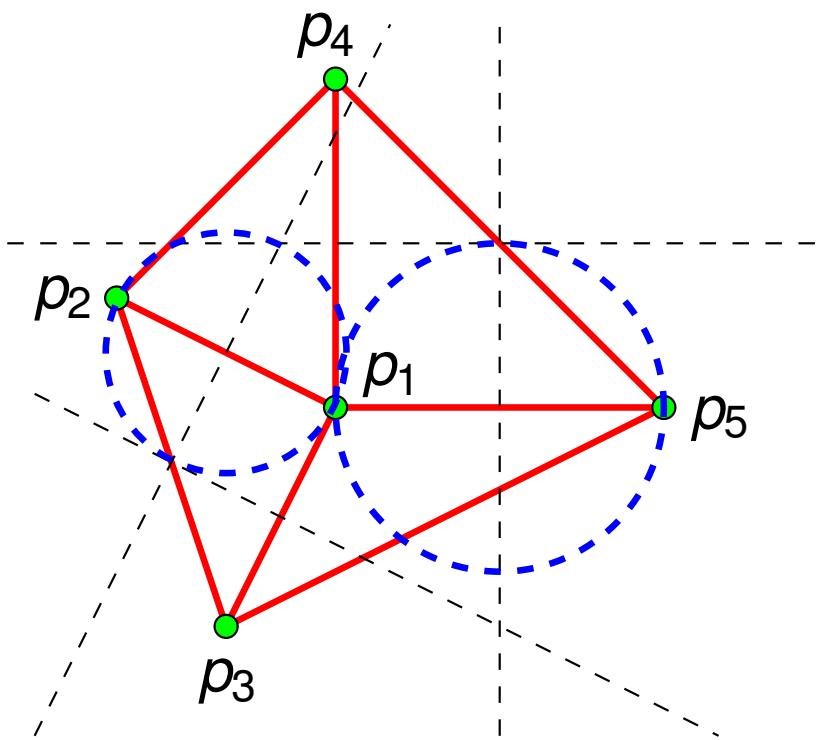
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the Delaunay triangulation

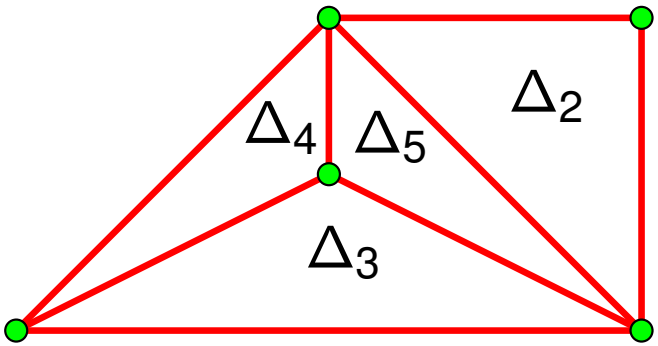


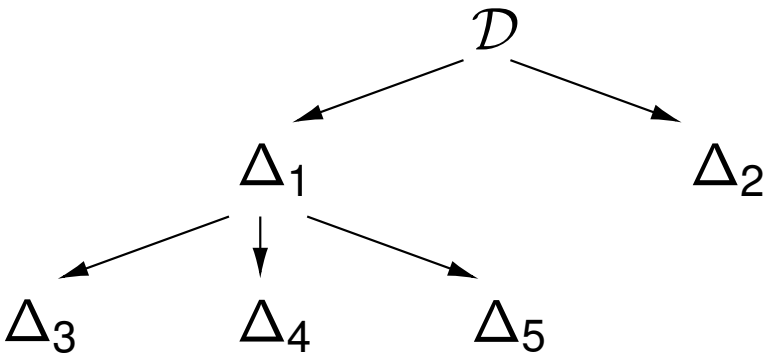
computing Delaunay triangulations

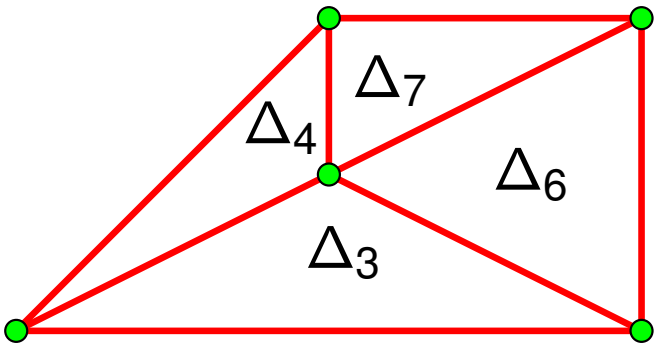


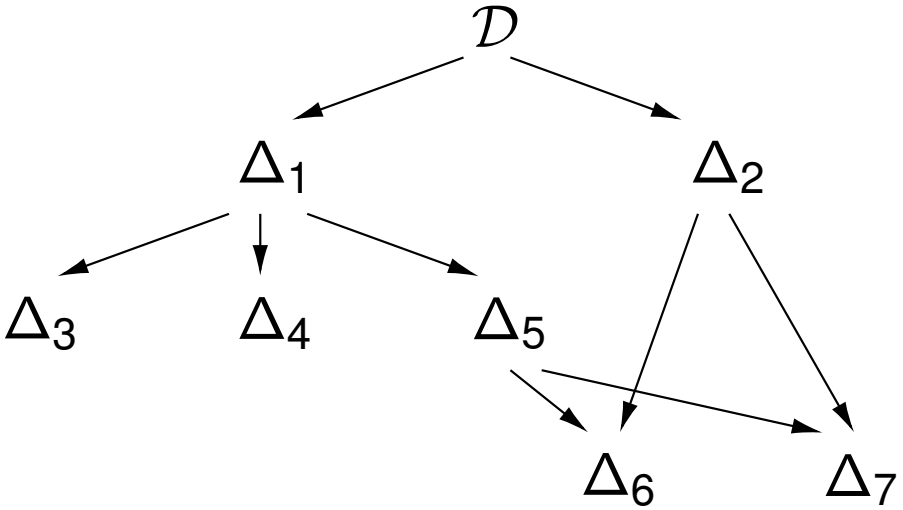


computing cost of Delaunay triangulations









4. circles in Delaunay triangulations

Consider a triangle $\Delta \in T$, in a Delaunay triangulation T.

Let $c = (c_x, c_y)$ be the center of the triangle Δ.

Consider any point $p \in \Delta$ and any point $q \in T \setminus \Delta$.

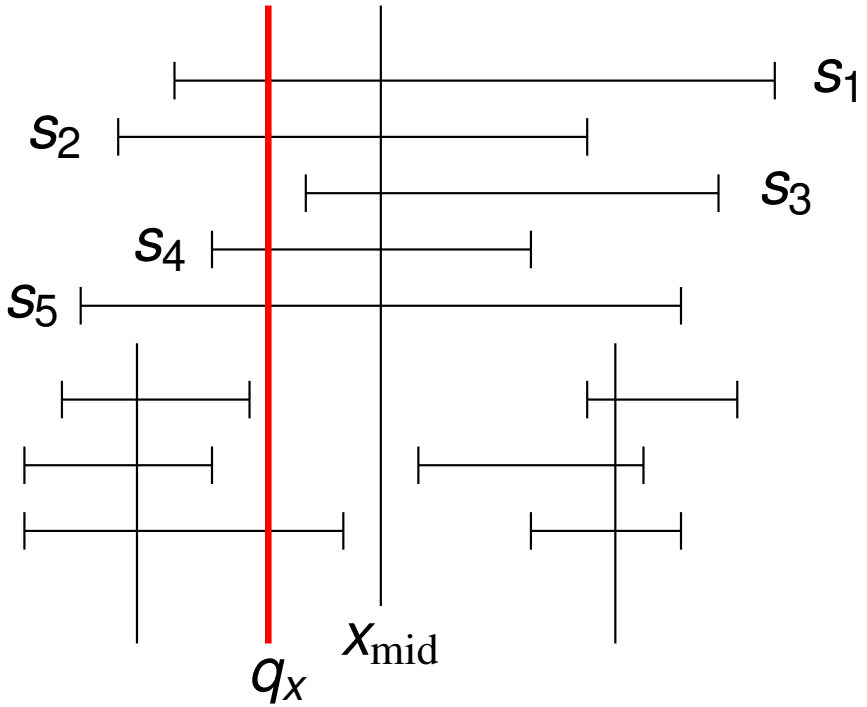
Denote by $d(p, c)$ the distance from p to c
and by $d(q, c)$ the distance from q to c.

How does $d(p, c)$ compare to $d(q, c)$? Justify.

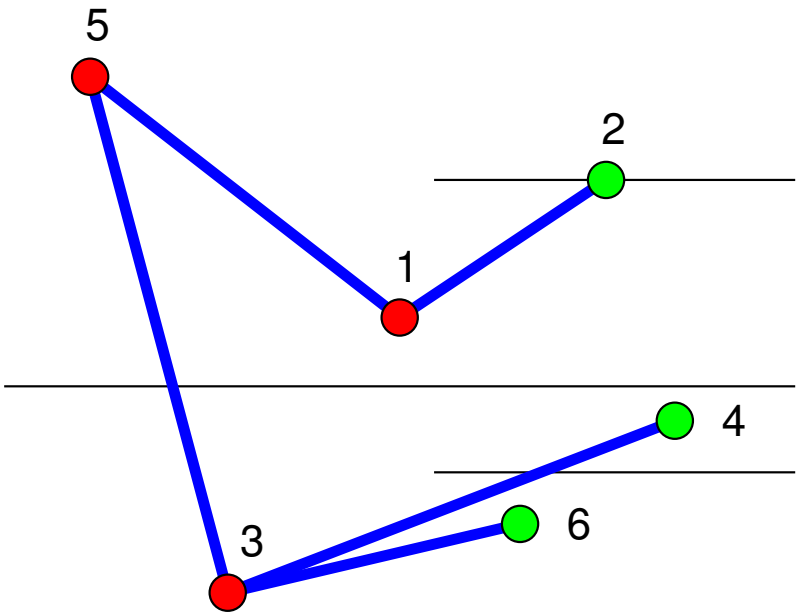
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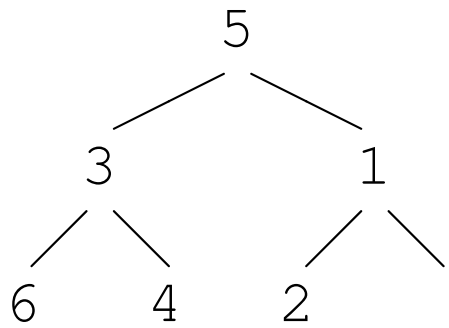
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interval trees

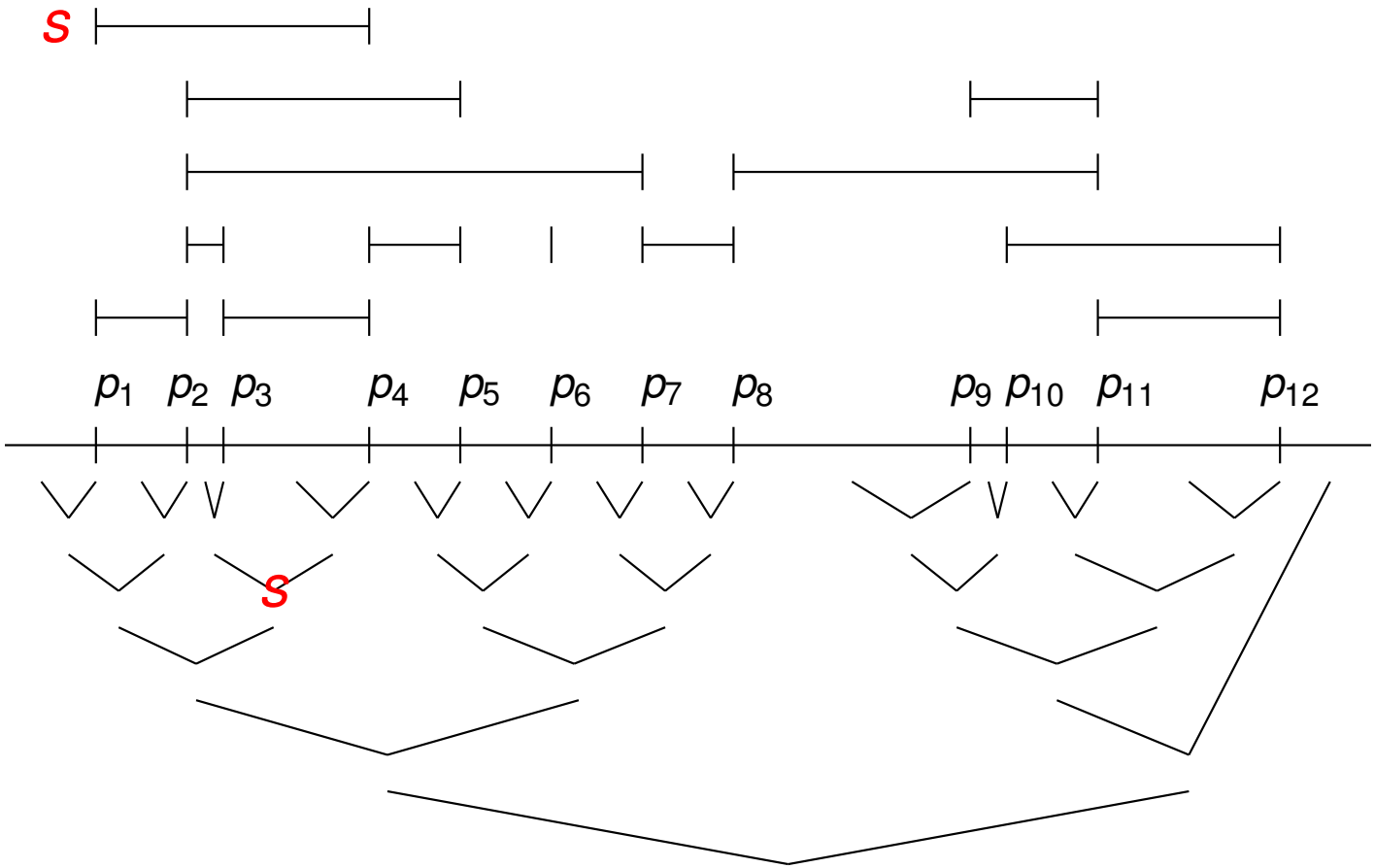


priority search trees





segment trees



5. data structures for windowing queries

Let I be the set

$$\{ [1:3], [2:8], [3:21], [5:21], [5:55], [13:89], [21:89], [34:55] \}.$$

Construct a segment tree to store I.

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1. Trapezoidal Maps

Consider the line segments

$$s_1(\underbrace{(0,1)}_{p_1},\underbrace{(1,3)}_{q_1}),\ s_2(\underbrace{(1,3)}_{p_2},\underbrace{(2,0)}_{q_2}),\ s_3(\underbrace{(0,1)}_{p_3},\underbrace{(2,0)}_{q_3}).$$

1. Draw the trapezoidal map and corresponding search structure obtained after inserting the segments in the order s_1, s_2, s_3.
2. How many faces are in this trapezoidal map?
3. Does this number stay the same for any three segments that define a triangle?

2. Voronoi diagrams

What is the largest degree of a vertex in a Voronoi diagram for n sites?

1. Illustrate your answer with an example of at least six sites.
2. Justify why your configuration of the sites leads to the largest degree.

3. arrangements of lines

Let $A(L)$ be a doubly connected edge list which stores an arrangement of n lines and the coordinates of a point p.

Assume p lies in the bounding box of the arrangement, that p is not a vertex and that p does not lie on any of the n lines.

Consider the problem of locating the face of $A(L)$ that contains p.

A primitive operation on a half edge and a point p returns left or right, depending on whether p lies to the left or to the right of the half edge.

1. What is the main idea for an efficient algorithm to locate the face of the arrangement that contains the point p.
2. What is the cost of such algorithm as a function of n?

4. Delaunay triangulations

Consider the points

$$a = (0,0),\ b = (2,0),\ c = (4,1),\ d = (3,1),\ e = (2,1),\ f = (1,1)$$

as input to the incremental algorithm to construct
a Delaunay triangulation.

1. Suppose the algorithm would not randomize and insert the points in the given order: a, b, c, d, e, f. Explain why this order is unfortunate for the cost.
2. What is the best order in which to insert the points? Justify your answer.

5. windowing queries

We defined the priority search tree of points for query windows of the type

$$W = (-\infty : q_x] \times [q_y : q'_y].$$

Consider query windows of the type

$$W = [q_x : q'_x] \times (-\infty : q'_y]$$

and illustrate the analogue priority search tree of at least six points.